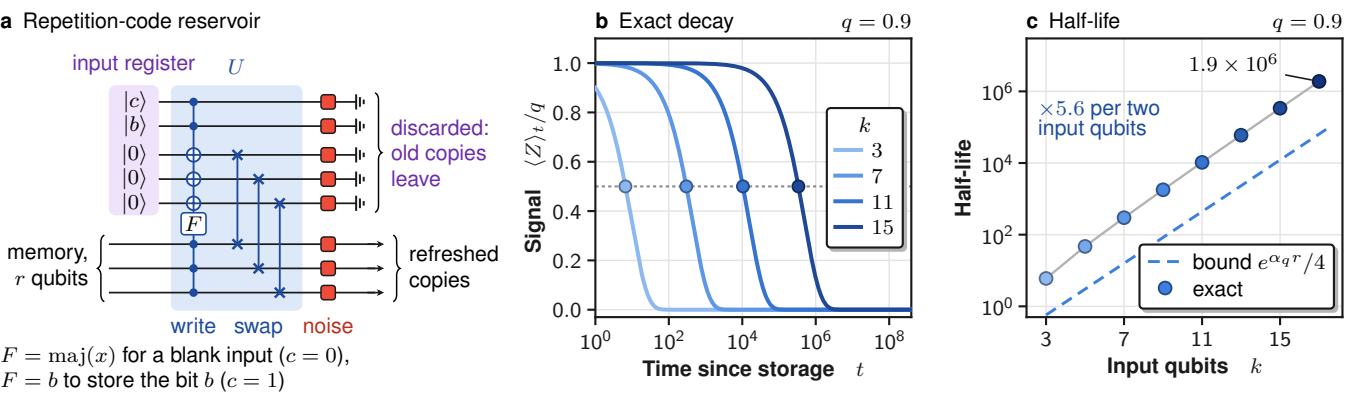

a Repetition-code reservoir
input register
$U$
$|c\rangle$
$|b\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$F$
discarded:
old copies
leave
memory,
$r$ qubits
refreshed
copies
write
swap
noise
$F = \mathrm{maj}(x)$ for a blank input ($c = 0$),
$F = b$ to store the bit $b$ ($c = 1$)
b Exact decay
$q = 0.9$
Signal $\langle Z \rangle_t / q$
1.0
0.8
0.6
0.4
0.2
0.0
$k$
3
7
11
15
$10^0$
$10^2$
$10^4$
$10^6$
$10^8$
Time since storage $t$
c Half-life
$q = 0.9$
$1.9 \times 10^6$
$\times 5.6$ per two
input qubits
Half-life
$10^6$
$10^4$
$10^2$
$10^0$
bound $e^{\alpha_q r}/4$
exact
3
7
11
15
Input qubits $k$